

SMOKE DETECTOR SYSTEM

Academic STEM Project Abstract Department of Electronics and Communication Engineering

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Abstract

Early and reliable detection of fire hazards is paramount in protecting human lives and minimizing property damage. This study presents the design, implementation, and performance evaluation of an IoT-Based Smoke Detection and Fire Alert System. Built upon the NodeMCU (ESP8266) microcontroller, the prototype integrates a highly sensitive MQ-2 metal-oxide semiconductor gas sensor to monitor concentrations of smoke and combustible gases (including LPG, carbon monoxide, propane, and hydrogen) in real-time. A 16x2 I2C Liquid Crystal Display (LCD) serves as a local interface to continuously display gas concentration metrics in raw analog levels.

When ambient gas levels exceed a calibrated safety threshold of 300 PPM, the system implements a dual-alert mechanism: engaging a high-decibel active piezoelectric buzzer and a high-brightness Red status LED, while switching the LCD display message to 'Smoke Detected!'. In critical conditions (readings > 750 PPM), the warning escalates to a high-priority 'WARNING! FIRE ALERT'. A secondary Green LED remains continuously illuminated under normal environmental conditions (readings below 300 PPM) to provide visual confirmation of system operational readiness and safe air quality. Laboratory testing indicates that the MQ-2 sensor exhibits high sensitivity and rapid response times (within milliseconds) to combustion by-products, making this system an effective and low-cost solution for residential and industrial fire safety monitoring. Future enhancements aim to incorporate IoT mesh networking, ESP-NOW point-to-point communication, and cloud dashboard integration (such as ThingSpeak or Blynk) to enable remote monitoring and push notifications to smartphones.